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Late Hercynian remagnetization of Tournaisian series from the Laval syncline, Armorican Massif, France

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Paleomagnetic investigations have been conducted on Tournaisian volcanics and sediments from the Laval syncline, in order to evaluate the consequences of the Late Variscan tectonic and thermic phases. On the southern flank of the syncline, anisotropy measurements have yielded maximum susceptibility in the schistosity planes. Thermal demagnetization exhibits only remagnetizations, characterized by a large range of blocking temperatures (from 350° to 670°C). Two phases of remagnetization seem to follow each other. The first may have occurred during the Tournaisian major orogenic phase. The second has taken place during the latest Stephanian/earliest Permian, in relation with the latest Variscan thermal event. All results are in favour of a clockwise rotation by 15–40° of the Laval syncline after the latest overprints. According to results from other regions of Brittany, this clockwise rotation involved the whole Armorican Massif. This motion may be related to the latest compression phase which has tightened the Ibero-Armorican arc, well before the Permo-Triassic opening of the Bay of Biscay.

1. Introduction

A series of paleomagnetic studies have been carried out on the Armorican Massif in order to evaluate the plate tectonic concepts applied to the Caledonian and Hercynian orogenies of Western Europe, by tracing the evolution of the directions of magnetization of the Cadomian block relative to the neighbouring blocks and microblocks [1–6]. Many of the investigated Eocambrian to Ordovician formations exhibit Carboniferous and Permo-Carboniferous overprints. The aim of this work is to identify those of the tectonic, magmatic or metamorphic events which could have been responsible for the remagnetizations and to determine possible late Variscan rotations in relation to those events. The syncline of Laval has been chosen because of the presence of early Carboniferous volcanic series which have recorded all late Variscan orogenic phases.

2. Geological setting and sampling

The Carboniferous basin of Laval is situated in the eastern part of the Median Synclinorium (Armorican central zone), which disappears eastwards under the Mesozoic layers of the Paris Basin (Fig. 1). To the west the outcrops of the Carboniferous formations are limited because of their perisynclinal geometry. Beyond that, they appear in strips and then are again well represented in the Menez Belair. The cartographic discordance of the lower Carboniferous on the Ordovician, Silurian and lower Devonian series, as well as the molasse sedimentation of the Tournaisian deposits of the Huisserie Formation [7–9] and the associated acid volcanism, all provide evidence of the Bretonic phase. During the upper Tournaisian, infralittoral calcareous sedimentation replaced the detritic sedimentation and continued into the Visean (Laval and Changé limestones). From latest mid-

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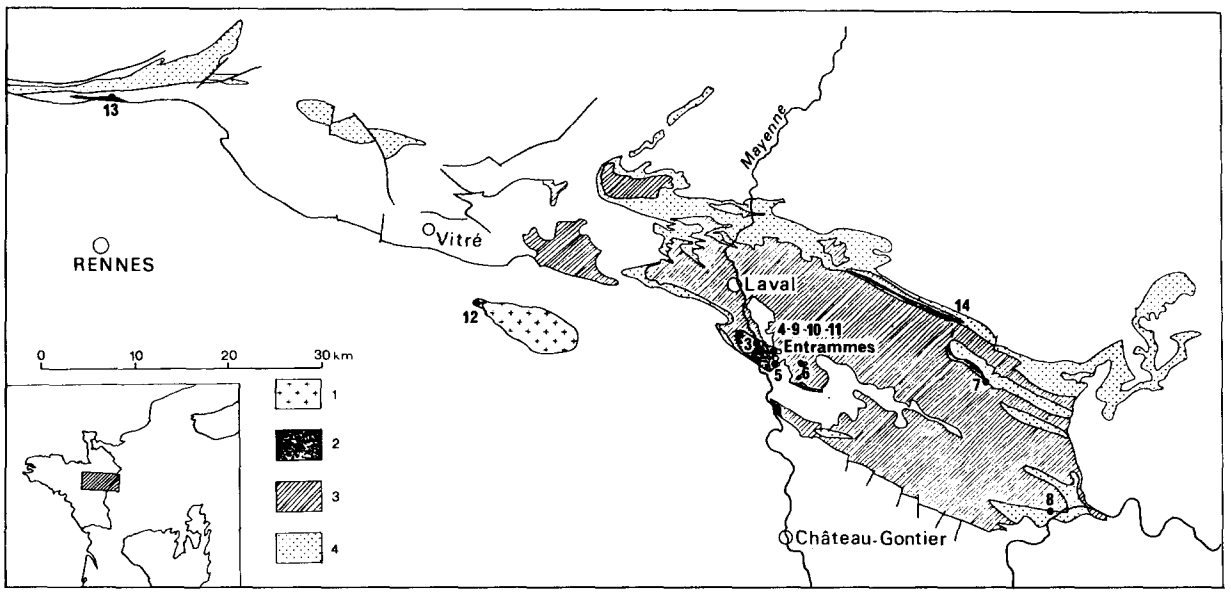


Fig. 1. Geological setting and site location: *1* = granite; *2* = Tournaisian volcanics; *3* = Carboniferous; *4* = Devonian.

dle Visean to early Westphalian (Laval shales) times, tectonic events and volcanic eruptions gave rise to a new detritic sedimentation. These events are equivalent to the Sudetic-1 phase of eastern France [10]. The formations are folded in an Armorican synclinal structure by a late early Westphalian tectonic phase (Erzgebirge), with upright or steeply inclined folds. Schistosity is more intense on the southern side of the syncline and is associated with recrystallization in the acid volcanics (rhyolites and ignimbrites). Finally, synsedimentary faulting affected deposition during Westphalian and Stephanian times. Paleomagnetic sampling has been concentrated on the Tournaisian volcanics of the Laval syncline, studied by Boyer in 1974 [11] and Plaine in 1976 [9]. In the Entrammes area (southern side) we sampled pale ignimbrites with "fiame" structures (sites 3, 5), layered ignimbritic rhyolites (sites 10, 11) and keratophyres (sites 4, 6, 9). The number of good outcrops is limited on the northern side (sites 7, 14). In the Sablé area, we sampled Tournaisian formations (sandstones with shales and antracitic seams) overlying lower Devonian limestones (site 8). West of the Laval syncline, two sites were in the rhyolites of Argentré du Plessis (site 12) and the Tournaisian rhyolitic tuffs of Saint-Germain sur Ille (site 13) north of Rennes (Menez Belair).

3. Microscopic investigations

Microscopic investigations of the opaque minerals provide interesting data for this paleomagnetic study. The rhyolites and ignimbrites generally show a complete alteration of primary opaque minerals in the southern limb of the syncline of Laval, and a partial alteration in the northern limb. Only pyrite remains unaltered. The weathered rocks contain rutile. The alteration of the opaque minerals is less in the keratophyres. At site 9 are fresh opaque minerals (titanomagnetite, ilmenite, rare bornite surrounded with a covellite halo and hematite in cracks). The differential alteration between ignimbrites and keratophyres is in part related to the type of mesostasis of these rocks (resistant microlitic structure in keratophyres, brittle glass structure in rhyolites and ignimbrites). The relation between schistosity and alteration of the opaque minerals in the ignimbrites and rhyolites provides good evidence for late remagnetizations during the main Westphalian deformation

phase, although other remagnetizations are also possible in relation to later thermal and tectonic events.

4. Sampling and laboratory techniques

All samples have been collected as handsamples. Orientation with magnetic and, when possible, solar compass, was recorded on a plaster cap. Samples were drilled and cut into specimens of 2.5 cm in diameter and 2.5 or 2.2 cm in length.

Remanent magnetizations have been measured with a Digico magnetometer with a noise level around 3×10^{-5} A m^{-1}. Susceptibilities have been measured with a Digico susceptibility bridge, and anisotropy of susceptibility with an anisotropy delineator of Digico. Alternating field and thermal demagnetization units have been built in our laboratory.

5. Normal remanent magnetization and susceptibilities

The highest intensities of normal remanent magnetization (NRM) appear in keratophyres (sites 6, 9) and in ignimbrites (site 7), in which the values reach 10^{-2} to 10^{-1} A m^{-1}. The other volcanics exhibit intensities of 10^{-4} to 2×10^{-3} A m^{-1}. In the sediments the values scatter around 10^{-3} A m^{-1}. Susceptibilities show the same tendency. The highest values of 3×10^{-1} (SI), have been obtained in keratophyres (site 9) and the weakest, 10^{-5}, in rhyolites and ignimbrites (site 10, 3).

6. Anisotropy of susceptibility

Anisotropy measurements have been performed to evaluate the consequences of the post-Namurian schistosity on ferromagnetic minerals. A significant anisotropy appears in four sites from the region of Entrammes in the southern part of the syncline (sites 4, 5, 6, 11; Fig. 2). Susceptibility is highest in sub-vertical planes oriented N120–130°, which coincide with the schistosity plane observed

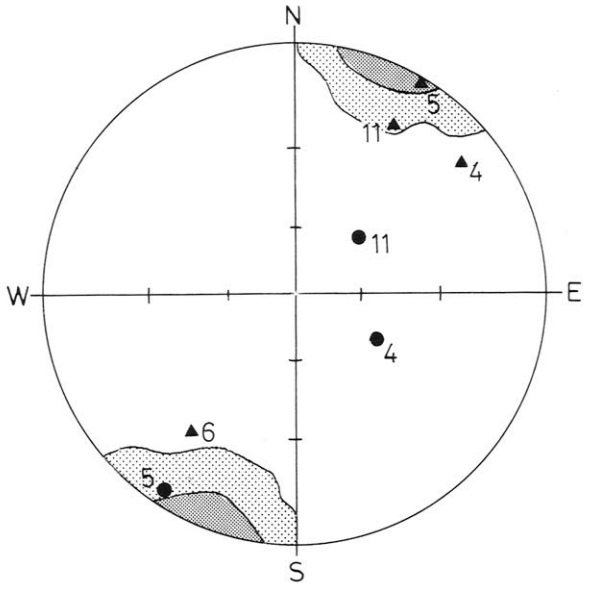


Fig. 2. Directions of minimum susceptibility (triangles) (susceptibility is maximum along orthogonal axes) and directions orthogonal to the bedding planes (dots) of sites 4, 5, 6, 11. Statistical representation of the schistosity poles (5 and 20%) after Plaine [9].

by Plaine [9]. In the ignimbrite of site 5, which is the most external of the southern limb, the plane of maximum susceptibility and the plane of schistosity observed in the samples coincide with the stratification given by the "fiames". Here the mean magnitude of anisotropy is 15% while in the other sites it scatters around 5%. Anisotropy exists probably also in sites 3 and 10 but measurements are not significant because of the weakness of the susceptibility.

7. Remanent magnetizations

Demagnetizing tests have shown that heating isolates better the different components of NRM than AF processing. This is due to rather high coercitivities (> 25 mT) and because of the appearance of anhysteretic magnetizations above 25 mT. Consequently, all specimens were demagnetized up to 600°C, and when necessary, up to 670°C. In all sites, demagnetization exhibits an

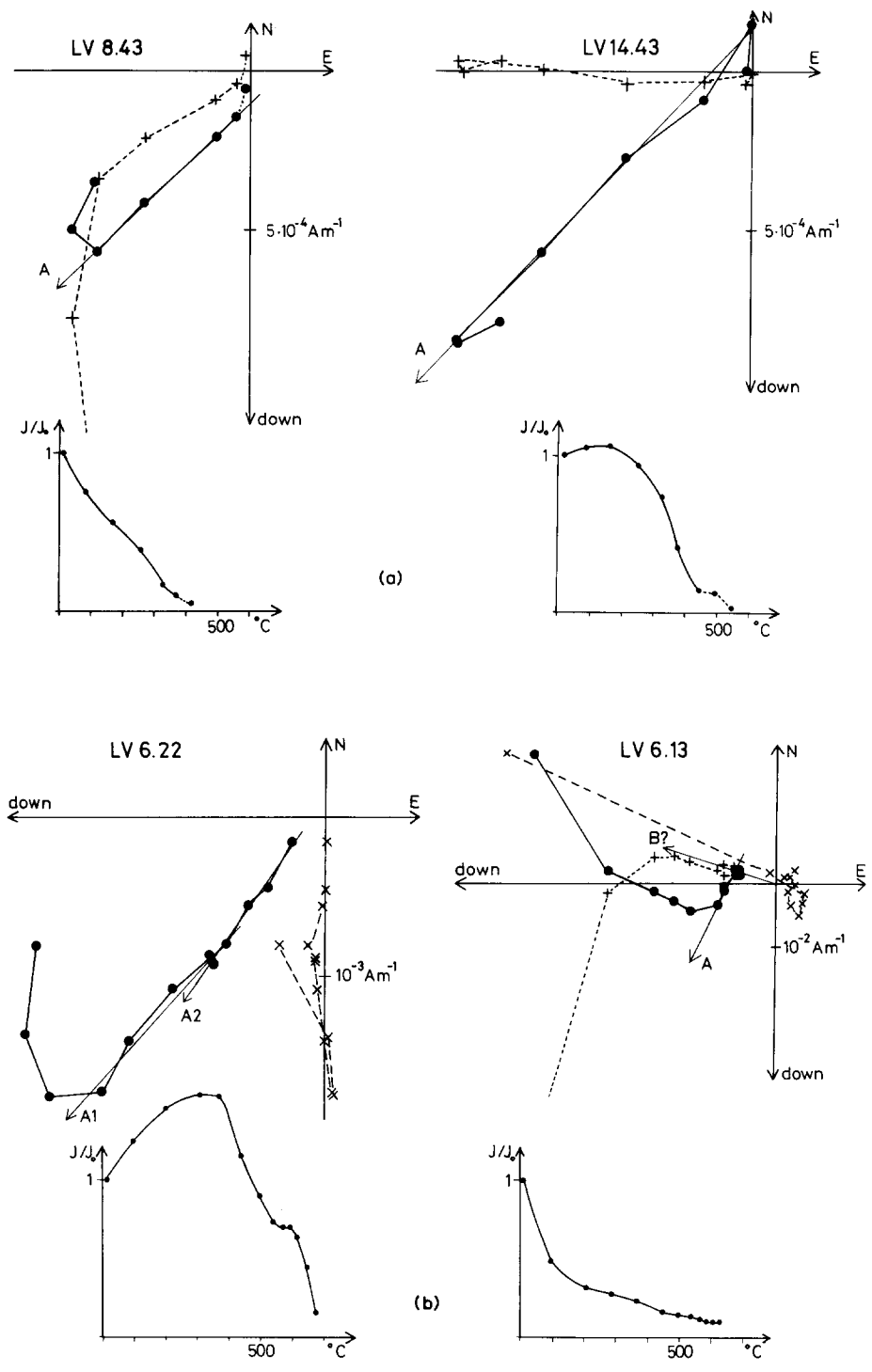


Fig. 3. Typical thermal demagnetizations. Dots represent orthogonal projections in the horizontal plane, crosses represent projections in the vertical E-W plane (+) or N-S plane (×).

TABLE 1

Site means of the characteristic directions of magnetization

Site	Formation		T_b (°C)	N/N_0	n	D (°)	I (°)	Dip (°)	d_c (°)	I_c (°)	k	α_{95} (°)
LV3	ignimbrite	Tournaisian	?	1/4	4	112	−51	55N340	132	−6	34	16
			?	1/4	2	258	−19	55N340	238	−17		
			?	1/4	2	203	10	55N340	228	43		
LV4	keratophyre	Tournaisian	500–580	1/4	3	194	−26	40N300?	181	−10	122	11
			> 580	1/4	4	312	−57		79	−80	32	16
LV5	ignimbrite	Tournaisian	350 ?	3/4	9	208	−11	87N 35	189	74	33	9
LV6	keratophyre	Tournaisian	580, 670	4/5	12	202	−25	?			19	10
			> 580	5/5	11	304	9				14	13
LV7	ignimbrite	Tournaisian	580	6/6	7	243	−8	60N190?	267	−36	112	6
				4/6	4	222	−16		260	−58	23	20
LV8	sandstones limestones	L. Tournaisian E. Devonian	350–400	6/6	10	224	25	65N 35	307	82	113	5
LV9	trachyte	Tournaisian	580	2/2	6	219	−12	?			123	6
			> 580	2/2	6	306	22				27	13
LV10	rhyolite	Tournaisian	300–500	3/3	4	205	−18	20N 60	207	−1	19	21.5
LV11	rhyolite	Tournaisian	350 ?	3/3	6	206	−6	37N230	198	−39	20	15.5
LV12	rhyolite	?	580	1/4	2	179	−34	?				
			500 ?	1/4	2	216	18					
			350–400	2/4	3	242	−13					
LV13	tuff	Tournaisian	350–400	2/2	3	217	19	52N180	218	−23	55	17
LV14	rhyolite	Tournaisian	450–500	5/5	6	233	−2	70N215	261	−65	76	8

T_b = blocking temperatures; N/N_0 = ratio of the samples used for computation to the total number of samples analysed; n = number of specimens used for computation; D, I = mean directions before tilt correction (if underlined, it is a post-tectonic remagnetization); D_c, I_c = mean direction after tilt correction; k, α = Fisher statistic parameters.

A component characterized by a southwesterly declination and a shallow, positive or negative, inclination (Fig. 3). Unfolding of the syncline results in diverging directions of this magnetization (Table 1), probably also for sites in which we could not find any tilt criteria: sites 6 and 9, which are close to sites 4, 5, 10, 11, show very similar mean directions. Consequently, we conclude that the A component has been acquired during or after folding.

Blocking temperatures differ from one site to another. In sediments (site 8; Fig. 3, specimen LV8.43) and trachytic tufs (site 13), the A component disappears below 400°C. Above 400°C, the magnetization as well as the susceptibility increase, which is an indication of transformations of the ferromagnetic minerals. In sites 6 and 9 (keratophyre) and site 7 (ignimbrite) a clear blocking temperature of 580°C reveals magnetite according to microscopic investigations. Fresh titanomagnetites were observed in site 9. Sometimes magnetite coexists with haematite (670°C) (Fig. 3, specimen LV6.22). Both carry the same A component. In site 14 (rhyolite), demagnetization is complete below 500°C. The different blocking temperatures, related to the differential alteration of the opaque minerals observed with the microscope, probably represent different stages of oxidations due to the late Variscan tectonic and thermic phases.

In two sites (6 and 9), after demagnetization of the A component, we have measured a residual B magnetization, with intensities of 10^{-4} to 10^{-2} A m^{-1}, which becomes unstable above 580°C. Directions of that residual magnetization oscillate around a mean northwesterly direction with a shallow inclination (Fig. 3, specimen 6.13). In other sites (4, 5, 11) an instrumental effect due to the Digico magnetometer, resulting in northwesterly or southwesterly directions with shallow inclinations, could be proved for samples with low intensities of NRM ($< 3 \times 10^{-4}$ A m^{-1}) and for some

samples which show strong increasing of intensity above 350°C. As that remaining component cannot be demagnetized, we must consider it as not characteristic.

8. The late Variscan remagnetizations

If the B component has no reality as a primary magnetization, then all early Carboniferous series from Laval syncline have been remagnetized. The mean direction of the A component is $D = 220°$, $I = -6°$, with $k = 15$, $\alpha_{95} = 12°$ ($\lambda = 33°$N, $\phi = 129°$E) (Fig. 4b, direction 4). Scatter is much greater between sites than within the sites: k exceeds 50 in 6 sites. This must be explained by tectonic phenomena. Site means can be divided into three groups. The first, which corresponds to the volcanics around Entrammes, on the southern limb of the syncline, has a mean direction of $D = 210°$, $I = -15°$, with $k = 63$, $\alpha_{95} = 8.5°$ ($\lambda = 42°$N, $\phi = 137°$E). The second corresponds to volcanics from the northern limb of the syncline (sites 7, 14). Two samples of site 12 (rhyolite) show a similar mean direction: $D = 238°$, $I = -8°$ ($\lambda = 24°$N, $\phi = 112°$E), which is more westward than the previous one. The last group is represented by the rhyolitic tufs (site 13) of St. Germain-sur-Ille, north of Rennes, and by the late Devonian early Tournaisian sediments (site 8) from the eastern part of the syncline. The mean direction is $D = 220°$, $I = 22°$ ($\lambda = 21°$N, $\phi = 137°$E).

Compared to directions computed from early Carboniferous [12], late Carboniferous/early Permian and late Permian [13] poles for stable Europe (British isles, northern and central Europe), all three groups of directions support a clockwise rotation of the Laval Basin after the time the remagnetizations took place. Remains the question of the age of these remagnetizations and consequently the tectonic phases related to the rotation. The mean inclination of the first group is comparable to that computed for the late Permian pole of Europe (Fig. 4b). In the Vosges and Morvan

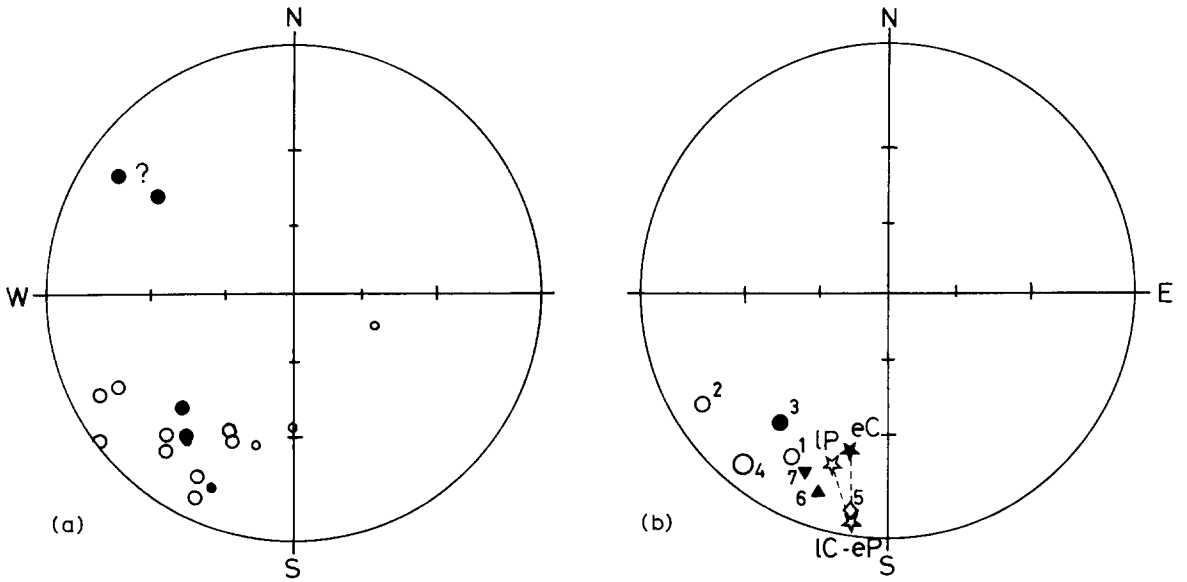


Fig. 4. (a) Stereographic projections of all in situ site means directions of magnetizations measured in the Laval syncline series. Small symbols represent mean directions from individual samples. (b) Stereographic projection of mean directions of: (*1*) the first group: Stephano-Permian remagnetizations in the Entrammes region; (*2*) the second group: Stephano – Permian remagnetizations on the north limb of the Laval syncline; (*3*) the third group: Westphalian remagnetizations; (*4*) the three groups together; (*5*) late Variscan remagnetizations in Vosges and Morvan [14]; (*6*) the Ploumanac'h granite (303 Ma) [2]; (*7*) the Flamanville granite [1]. Stars represent the directions compiled for the Laval syncline, from the early Carboniferous [12], late Carboniferous/early Permian and late Permian [13] virtual geomagnetic poles.

[14], late Variscan magnetizations are consistent with the late Carboniferous/early Permian mean direction of Europe. This later direction is derived mainly from acidic volcanic series dated at 290–280 Ma, i.e. the late Carboniferous/early Permian boundary. Either the remagnetizations of the first group have taken place during the late Permian and the basin was rotated later, or they are contemporaneous with those observed in other Variscan regions. The possibility of partial tilt remagnetization which occurred during the end of folding cannot be excluded. In that case, the inclination would be less important or even positive. The high blocking temperature of magnetite (580°C) measured on sites 6, 7 and 9 is in favour of a deuteric setting of the overprints. Because in the Armorican Massif the thermal events had terminated before 290 Ma [15], the hypothesis of latest Carboniferous/earliest Permian remagnetizations appears the most probable.

The second group, which represents the northern limb of the syncline, yields inclinations consistent with those observed for remagnetizations from the Vosges and Morvan, while the declinations are in favour of a 40° clockwise rotation. In both cases, for the first and the second group, remagnetizations were emplaced during the reversed Kiaman interval, probably in the 300–290 Ma timespan, and not as one would have expected it, during the Westphalian tectonic phase which has induced the schistosity.

The positive inclination of the third group, comparable to that computed from the early Carboniferous pole of stable Europe, is positively in favour of an age for that remagnetization which is older than that of the former groups. In site 8, tilt correction yields an unrealistic high positive inclination. This indicates that the remagnetization has taken place after folding, i.e. after the earliest Westphalian. According to Fourmarier and Pelhate [16] and Plaine [9], development of schistosity must have resulted from a syntectonic granitic phase, somewhere beneath the Laval syncline. The thermal conditions favoured recrystallization in the schistosity planes. In the northern Armorican Massif, a series of Westphalo-Stephanian granites was emplaced which cooled in the 305–290 Ma timespan: the granites of Ploumanac'h (303 Ma [15]), Flamanville, Quintin (291 Ma [17]), etc.

Paleomagnetic results from the Ploumanac'h and Flamanville granites [1,2] show inclinations comparable, but a little shallower than those for the third group (Fig. 4b, sites 3, 6, 7). Consequently, a Westphalian age of 310–300 Ma, corresponding to the hydrothermal activity induced by the granitization, seems to be reasonable for the earliest remagnetization. Blocking temperatures of 350–400°C show that for both sites (13 and 8), geotherms have remained under 400°C during the Westphalo-Stephanian while in the Entrammes region they may have surpassed 500°C.

9. Tectonic implications

Since the Westphalian and Stephano-Autunian, directions of remagnetizations deviate by 15–40° relative to the direction computed from the corresponding poles of Europe, a clockwise rotation must have occurred after acquisition of the latest overprint, i.e., during and/or after latest Stephanian. That rotation appears also in other regions from the Armorican Massif. Directions of Carboniferous remagnetizations around N210° were measured on so-called Precambrian/Cambrian diorites [5], Cambro-Ordovician red beds [4] and Ordovician dolerites [6] from northern and western Brittany. The directions from the Ploumanac'h and Flamanville granites which are located in the northern Armorican Massif show also a similar but weaker deviation. The general clockwise motion of the Armorican Massif, to which can be related late Variscan dextral strike-slip faulting [18–20], may be explained by the latest compression phase between the Iberian nucleus and Laurussia, that tightened the Ibero-Armorican arc [21,22].

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